

Interview Prep

Master Index & Study Companion

Version 1.0 — Covers all 7 PDFs in your collection

What's inside this index

- Complete catalog of all 7 PDFs in your prep collection
- One-line summary + key topics + sample Q&A from each
- Study workflows for different timelines (1 week / 2 weeks / 1 month)
- How to use the PDFs together — what to read in what order
- Combined master cheatsheet — the best of all 7 PDFs in one place
- Interview-day checklist

PYTHON

DEVOPS

1. The 7 PDFs at a Glance

#	PDF	Type	Content	Pages
1	devops_lead_interview_questions.pdf	DevOps	50 questions, 9 topics	~15
2	devops_lead_interview_questions_part2.pdf	DevOps	50 advanced questions (Argo CD, GitHub, CloudBees, Runners, EKS)	~14
3	UsefulBI_DevOps_Interview_QA_Workflows.pdf	DevOps	94 questions + workflows + system designs + behavioral	22
4	DevOps_Interview_Master_Skill_v1_0.pdf	DevOps	~200+ questions across Beginner/Intermediate/Advanced + cheatsheets	40
5	python_interview_mastery_v1.0.pdf	Python	133 questions, 15 topics, 5 cheatsheets	55
6	python_leetcode_top150_v2.0.pdf	Python	124 LeetCode problems, 14 patterns, 5 cheatsheets	67
7	python_advanced_book2_v2.0.pdf	Python	71 questions: Async + ML/Data + Django/FastAPI + cheatsheets	30

Totals: ~672 questions/problems across 7 PDFs · ~243 pages total · 19 cheatsheets

2. PDF Details & Sample Q&A

#1 DevOps Lead Interview Questions

DEVOPS

File: [devops_lead_interview_questions.pdf](#)

SUMMARY

50 questions covering the breadth of a DevOps Lead role across 9 topics with answers and color-coded sections.

TOPICS COVERED

CI/CD · Containers · Infrastructure · Monitoring · Security · Cloud · Scripting · Culture/Process · Leadership

BEST FOR

First-pass review of DevOps fundamentals; quick recall before phone screens.

SAMPLE Q&A

Q: How do you handle a production deployment that's failing?

A: Stop the deployment, assess impact (rollback vs forward-fix), communicate status, check logs and recent changes, rollback if user-facing impact is high, then root-cause after recovery. Write postmortem with action items.

#2 DevOps Lead Interview Questions — Part 2 (Advanced)

DEVOPS

File: [devops_lead_interview_questions_part2.pdf](#)

SUMMARY

50 deeply technical questions across 5 specific tool areas — 10 questions each.

TOPICS COVERED

Argo CD · GitHub · CloudBees · Runners · EKS

BEST FOR

Deep tool-specific drilldown; UsefulBI / Cisco / FAANG style tool questions.

SAMPLE Q&A

Q: Explain Argo CD app-of-apps and when you'd use ApplicationSet instead.

A: App-of-apps: a parent Application manages child Applications, useful for bootstrapping a cluster declaratively (root → platform-addons → shared-services → business-apps). ApplicationSet: generates many Applications from a template using generators (list, cluster, Git, matrix, PR). Use ApplicationSet when deploying the same app across many clusters or environments.

#3 UsefulBI DevOps Interview Q&A Pack

DEVOPS

File: *UsefulBI_DevOps_Interview_QA_Workflows_SystemDesign_v1_0.pdf*

SUMMARY

Targeted at UsefulBI / FAANG / Cisco / Meta / Palo Alto / Zscaler interview style. 94 Q&A with workflows, system designs, and behavioral answers.

TOPICS COVERED

AWS · Terraform · Docker · Kubernetes/EKS · Helm · CI/CD · GitOps · Databricks/MLOps · Observability · Security · Behavioral

BEST FOR

Bay Area cloud/data platform interviews; companies that want system design + AWS + EKS depth.

SAMPLE Q&A

Q: How would you design a production-grade EKS platform for multiple teams?

A: Layered design: separate AWS accounts per env, private EKS clusters, Karpenter for node autoscaling, IRSA for pod identity, External Secrets, cert-manager, AWS Load Balancer Controller, observability stack (Prometheus/Grafana), network policies, admission policies, standardized Helm charts, tenant namespaces with RBAC + resource quotas. Use GitOps (Argo CD) for delivery.

#4 DevOps Interview Master Skill

DEVOPS

File: *DevOps_Interview_Master_Skill_v1_0.pdf*

SUMMARY

The biggest DevOps PDF. ~200+ questions organized as Beginner / Intermediate / Advanced with deep topic coverage, troubleshooting workflows, and final cheatsheets.

TOPICS COVERED

Linux · Networking · DNS/TLS · Docker · Kubernetes · EKS · Helm · Argo CD/GitOps · Terraform · AWS · Jenkins/CloudBees · Security · Observability · SRE · Behavioral · Scenario rounds

BEST FOR

Comprehensive study — go here for any topic you need deep coverage on. Includes scenario rounds (Meta, Cisco, Palo Alto, Zscaler, Zoom, Oracle, AppZen, startup, FAANG).

SAMPLE Q&A

Q: Why can CPU limits hurt latency in Kubernetes?

A: CPU limits can cause CFS throttling even when node CPU is available, increasing tail latency. Many production systems set CPU requests but avoid strict CPU limits for latency-sensitive workloads.

#5 Python Interview Mastery (v1.0)

PYTHON

File: [python_interview_mastery_v1.0.pdf](#)

SUMMARY

Broad Python interview prep. 133 questions across 15 topics with Beginner/Intermediate/Advanced color-coded tiers.

TOPICS COVERED

Fundamentals · Data Structures · Algorithms · OOP · Decorators/Generators · Concurrency · Internals · Exceptions · Strings/Regex · FAANG problems · Pythonic Idioms · System Design · Testing · Stdlib · Company-Favorites

BEST FOR

First pass for Python interviews; general breadth coverage before specializing.

SAMPLE Q&A

Q: What's the difference between `__init__`, `__new__`, and `__call__`?

A: `__new__` creates the instance (returns it). `__init__` initializes the already-created instance (no return). `__call__` makes the instance itself callable like a function. Order: `__new__` → `__init__` for instance creation; `__call__` is separate, invoked when you call the object.

#6 Python LeetCode Top 150 (v2.0 Book 1)

PYTHON

File: [python_leetcode_top150_v2.0.pdf](#)

SUMMARY

Deep coding patterns book. 124 LeetCode problems organized by 14 recurring patterns. Each solution has problem statement, approach, complexity, Python code, key insight.

TOPICS COVERED

Arrays/Strings · Two Pointers · Sliding Window · HashMap · Linked List · Stack · Trees/BST · Trie · Graphs · Heap · Backtracking · DP · Binary Search · Math/Bit

BEST FOR

FAANG-style coding rounds. The 'pattern recognition' lens — see the pattern, write the code.

SAMPLE Q&A

Q: #42 Trapping Rain Water (Hard)

A: Two pointers. Track `max_left` and `max_right`; water at each position = $\min(\text{max_left}, \text{max_right}) - \text{height}[i]$. Move the shorter side inward. $O(n)$ time, $O(1)$ space. Key insight: water level is bounded by the shorter of the two max walls seen so far.

File: [python_advanced_book2_v2.0.pdf](#)

SUMMARY

Specialty deep-dive across 3 domains: Async/Concurrency (19 Qs), ML/Data Engineering (25 Qs), Django/FastAPI (27 Qs). With code examples and senior-engineer commentary.

TOPICS COVERED

Asyncio · Coroutines · Event Loop · TaskGroup · uvloop · NumPy/Pandas · scikit-learn · Feature Engineering · MLOps · ORM · DRF · Pydantic v2 · Dependency Injection · Async SQLAlchemy

BEST FOR

Senior Python interviews where you need to go deep on one or more specialty areas. Great for ML platform / web API roles.

SAMPLE Q&A

Q: Why does `asyncio.run(main())` fail if you call it from inside an already-running event loop?

A: `asyncio.run()` creates a new event loop, sets it as current, runs the coroutine to completion, and closes the loop. You can't have two running loops in the same thread — calling it inside an existing loop raises `RuntimeError`. Inside an async context, just use `'await main()'` directly.

3. Study Workflows by Timeline

Pick the timeline that matches when your interview is. Spend less time reading, more time speaking answers out loud.

■ 1-Week Crash Course (interview in <7 days)

Day	Focus	PDFs to Use
Mon	Python fundamentals + OOP	#5 (PDF 5) — chapters 1-4
Tue	DevOps fundamentals + AWS + K8s basics	#1 + #4 Beginner section
Wed	Coding practice — most common LC patterns	#6 — Arrays, Two Pointers, Sliding Window, Hash Map
Thu	Tool-specific drilldown (whichever apply)	#2 + #3 (system design)
Fri	Async OR ML OR Web (your specialty)	#7 — pick ONE section
Sat	Mock interview — speak answers OUT LOUD	Any 3 PDFs at random
Sun	Cheatsheet review only — DON'T learn new things	Cheatsheets from all PDFs

■ 2-Week Balanced Plan (interview in 2 weeks)

Week	Phase	PDFs	What to Do
1 Mon-Wed	Foundation	#5, #1	Read entire PDF #5 (Python) and PDF #1 (DevOps). Highlight what you do
1 Thu-Fri	Coding ramp	#6	Solve 2-3 problems per day FROM MEMORY (look up only after attempting
1 Sat-Sun	Deep DevOps	#3, #4	Read PDF #3 for system designs. Skim PDF #4 for advanced sections you
2 Mon-Tue	Tool deep-dive	#2	If they use Argo CD / EKS / GitHub Actions, this is your gold.
2 Wed-Thu	Specialty	#7	Pick the specialty section that matches the role (Async, ML, or Web).
2 Fri	Mock & Behavioral	#3 (behavioral), #4 (behavioral)	STORY format answers. Speak out loud or with a friend.
2 Sat	Cheatsheet review	All	Skim every cheatsheet. Don't learn new content.
2 Sun	Rest. Light review only.	—	Sleep. Hydrate. Trust the prep.

■ 1-Month Thorough Plan (interview in 4 weeks)

Week	Theme	Daily Work
1	Python breadth	Read PDF #5 cover-to-cover (1-2 chapters/day). Practice answering 5 Qs out loud each day. End-of-w
2	DevOps breadth + system design	Read PDF #1 then PDF #4 (Beginner + Intermediate). Read PDF #3 for system designs. Sketch each
3	Coding + tool depth	PDF #6: solve 4-5 LC problems/day. Pick patterns weakest first. PDF #2: read all 5 tool sections. Build
4	Specialty + mocks + behavioral	PDF #7: deep-dive into your role's specialty. PDF #4 advanced + behavioral. Do 3+ mock interviews. P

4. How to Use the PDFs Together

By Role Type

Role You're Interviewing For	Read These PDFs in This Order
DevOps Engineer / SRE	#1 → #4 (Beginner/Intermediate) → #3 (system designs) → #2 (tool deep-dives) →
DevOps Lead / Staff	#3 → #4 Advanced → #2 → behavioral sections of #3 and #4
Cloud / Platform Engineer (AWS-heavy)	#3 (AWS-focused) → #4 → #2 (EKS section)
Python Backend Engineer	#5 → #7 (Django/FastAPI section) → #6 (LC patterns)
Python ML / Data Engineer	#5 → #7 (ML/Data section) → #6 (just Arrays, HashMap, DP)
Python Generalist (FAANG)	#5 → #6 (all 14 patterns) → #7 (Async section)
DevOps with Python scripting	#1 + #4 + #5 (just Fundamentals + Scripting sections)

Study Mechanics That Work

- 1. Active recall beats re-reading.** Close the PDF, ask yourself the question, answer out loud. Then check.
- 2. Speak answers out loud.** Interviewers test how you explain, not just what you know. Reading silently doesn't simulate this.
- 3. For coding: solve first, look second.** Even 10 minutes struggling teaches more than 30 minutes of reading the solution.
- 4. Pattern recognition for LC.** When you see a new problem, ask: 'Have I seen this pattern?' Get to where you classify problems in 30 seconds.
- 5. Build STAR stories.** Pick 5-7 real stories from your career. Practice telling each in 2 minutes. Reuse them across behavioral questions.
- 6. The 80/20 of cheatsheets.** The last 1-3 hours before interview = cheatsheet review ONLY. Don't try to learn new things at the door.
- 7. Mock interview = irreplaceable.** Even one mock with a friend or paid service surfaces gaps no amount of reading will.

5. Master Cheatsheet — Best of All 7

The most-used patterns from all 7 PDFs, condensed.

Python — Data Structures & Built-ins

Need	Use
Sort	<code>sorted(it, key=fn, reverse=True)</code>
Counter / freq map	<code>collections.Counter(it)</code>
Default dict	<code>collections.defaultdict(list)</code>
Stack / queue / deque	<code>list (stack), collections.deque (both ends)</code>
Min/max heap	<code>heapq (min-heap; negate for max)</code>
Top k elements	<code>heapq.nlargest(k, it), nsmallest(k, it)</code>
Binary search	<code>bisect.bisect_left, bisect.insort</code>
Cache function results	<code>functools.lru_cache</code> or <code>@cache</code>
Cartesian product / combos	<code>itertools.product / combinations / permutations</code>
Running sum	<code>itertools.accumulate(it)</code>

LeetCode — Recognize the Pattern in 30 Seconds

Problem Says...	Pattern
sorted array, find pair / palindrome	Two Pointers
contiguous subarray/substring property	Sliding Window
top k / kth largest / k closest	Heap (size k)
shortest path / level by level	BFS
all paths / count components / cycle	DFS or Union-Find
dependencies / build order	Topological Sort
all subsets / permutations / N-queens	Backtracking
min / max / count of ways	Dynamic Programming
matrix sorted rows/cols	Binary Search or staircase
prefix / autocomplete / word lookup	Trie
next greater / smaller element	Monotonic Stack

Complexity — What's Acceptable at Each Input Size

$n \leq$	Acceptable Complexity
10	$O(n!)$ — permutations
20	$O(2^n)$ — subsets
500	$O(n^3)$
5,000	$O(n^2)$

10■	$O(n \log n)$ or $O(n)$
10■+	$O(n)$ or $O(\log n)$

Kubernetes — Debug Anything in 3 Steps

Symptom	Cause	Fix Path
Pod Pending	Scheduler can't place pod	describe pod → events → resources/taints/PVC/quotas
CrashLoopBackOff	App starts then dies	logs --previous → exit code → env/config/secrets → probes
ImagePullBackOff	Registry/auth/network	describe → image exists? → IAM/secret → NAT/endpoints
Service has no endpoints	Selector mismatch or no ready pods	Check selector labels + readiness probe
503 from ALB but pods healthy	Target group / readiness gate	ALB target group health → readiness probe path
Argo CD OutOfSync	Git ≠ live	argocd app diff → Git revert OR adjust ignoreDifferences

AWS Networking — Memory Tricks

AWS Term	Memory Hook
VPC	neighborhood
Subnet	street
Route Table	GPS route
Internet Gateway (IGW)	public exit
NAT Gateway	private outbound exit
Security Group	instance firewall (stateful)
NACL	subnet firewall (stateless)
VPC Endpoint	private road to AWS service
ALB	Layer 7 — understands URLs
NLB	Layer 4 — understands connections
IAM Role	who can do what
IRSA	pod identity for AWS

Answer Frameworks — Use These Verbatim

Question Type	Framework
Behavioral	STAR — Situation, Task, Action, Result
Troubleshooting	TROUBLE — Trigger, Reproduce, Observe, Unblock, Build fix, Learn, Eliminate recurrence
System Design	ARCH — Actors, Requirements, Components, High availability, Risks/tradeoffs
Cloud Security	SECURE — Scope, Encrypt, Control access, Update/patch, Rotate secrets, Evidence/logging
Production Incident	Detect → Triage → Mitigate → Communicate → Resolve → Postmortem
Tool Comparison	What it does → strengths → weaknesses → when I'd use it

6. Interview Day Checklist

Night Before

- Cheatsheet review only. Do NOT learn new things.
- Re-read your STAR stories — pick which one fits each behavioral question type.
- Lay out clothes, charger, water bottle, notepad, pen.
- Test your video setup and internet if remote.
- Look up the interviewer on LinkedIn (their background = clues about questions).
- Re-read the job description — note 3 keywords to bring up naturally.
- Sleep at least 7 hours. Caffeine doesn't replace this.

Morning Of (60 minutes before)

- Eat something — interview hunger is a confidence killer.
- Quick 5-min skim of THE ONE PDF most relevant to today's interview.
- Read your STAR stories one more time.
- Open a blank doc and brain-dump 5 facts in case you forget them mid-interview.
- Warm up your voice: explain something out loud (the more technical, the better).
- Have water and tissues nearby.
- 10 min before: stop reviewing. Stretch. Breathe deep 5x. Smile in the mirror.

During the Interview

- **Repeat the question back.** Buys thinking time AND confirms you understood.
- **Think out loud.** Interviewers grade your reasoning, not just your answer.
- **Don't apologize for thinking.** 'Let me think about that' > 'Sorry, I don't know.'
- **Ask clarifying questions.** Especially for system design — never assume requirements.
- **When stuck, say what you DO know.** Even partial answers earn partial credit.
- **For coding: edge cases first.** Empty input? Single element? Negative numbers? Duplicates?
- **Always have 2-3 questions ready for the interviewer.** About their work, the team, the role.
- **End strong.** 'I'm really excited about this role because...' — one sincere line.

After the Interview

- Write down questions you got — feed back into prep for the next round.
- Send a thank-you email within 24 hours. Mention something specific from the conversation.
- Don't ruminate. Whatever happens happens. Start prep for the next one.
- If rejected: politely ask for feedback. Half of recruiters will give it. That's gold.